

Oscillations

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Q1. A particle is executing simple harmonic motion. Its amplitude is A and time period is 5 sec. The time required by it to move from $x = A$ to $x = A/\sqrt{2}$ is ____ sec.

JEE Main 2026 (Online) 8th April Evening Shift

✓ **Answer: 0.625 (i.e. $5/8$ s)**

Solution:

Starting from the extreme position, $x(t) = A \cos(\omega t)$, with $\omega = 2\pi/T = 2\pi/5$.

Set $x = A/\sqrt{2}$: $\cos(\omega t) = 1/\sqrt{2}$, so $\omega t = \pi/4$.

$t = (\pi/4)/(2\pi/5) = 5/8 = 0.625$ s.

Q2. A spring stretches by 2 mm when it is loaded with a mass of 200 g. From the equilibrium position the mass is further pulled down by 2 mm and released. The frequency associated with the system and the maximum energy stored in the spring are ____ Hz and ____ J, respectively. (Take $g = 10 \text{ m/s}^2$)

JEE Main 2026 (Online) 6th April Evening Shift

✓ **Answer: $f \approx 11.25$ Hz, $U_{\max} = 8 \times 10^{-3}$ J**

Solution:

Spring constant: $k = mg/x_0 = (0.2)(10)/0.002 = 1000 \text{ N/m}$.

Angular frequency: $\omega = \sqrt{k/m} = \sqrt{(1000/0.2)} = \sqrt{5000} \approx 70.71 \text{ rad/s}$.

Frequency: $f = \omega/2\pi \approx 11.25 \text{ Hz}$.

At the lowest point the spring is stretched by $x_0 + A = 2+2 = 4 \text{ mm} = 0.004 \text{ m}$.

Maximum spring energy: $U_{\max} = \frac{1}{2}k(x_0+A)^2 = \frac{1}{2}(1000)(0.004)^2 = 8 \times 10^{-3} \text{ J}$.

Source: JEE Main 2026 (NTA official releases, physics section) · [formulaverse.in](https://www.formulaverse.in)